"INTRODUCTION TO ENERGY" WORKSHEET

Name_____

Date _____ Class_____

Part 1. The two basic types of energy

Directions: Determine the best match between basic types of energy and the description provided. Put the correct letter in the blank.

_____1. A skier at the top of the mountain (a) Kinetic Energy

_____2. Gasoline in a storage tank (b) Potential Energy

_____3. A race-car traveling at its maximum speed (c) Both forms of Energy

_____4. Water flowing from a waterfall before it hits the pond below

_____5. A spring in a pinball machine before it is released

_____6. Burning a match

_____7. A running refrigerator motor

Part 2. Definitions of Energy.

Directions: Write down the definition for each of the following terms.

ENERGY:

KINETIC ENERGY:

POTENTIAL ENERGY:

Part 3. Forms of Energy.

Directions: Determine the type of energy for each form (Kinetic, Potential, or Both) and give an example.

Form	**Definition**	**Type (KE, PE, or Both)**	**Example (for each type if both)**
Mechanical (motion) energy	An object's movement creates energy		
Thermal (heat) energy	The vibration and movement of molecules		
Radiant energy	Electromagnetic waves		
Electrical energy	Movement of electrons		

Chemical energy	Stored in bonds of atoms and molecules		
Nuclear energy	Stored in the nucleus of an atom; released when nucleus splits or combines		
Sound energy	Vibration of waves through material		
Gravitational energy	Energy of position or height		

Part 4. Forms of Energy Continued

Directions: Match the energy form(s) to the description provided. A few questions may have more than one answer.

_____1. Falling rocks from the top of a mountain (a) Mechanical

_____2. Release of energy from the Sun (b) Electrical

_____3. Energy released from food after it is eaten (c) Heat

_____4. Batteries (d) Radiant

_____5. The energy that runs a refrigerator (e) Chemical

_____6. Nuclear fission reactors (f) Nuclear

_____7. The rumble of thunder from a storm (g) Sound

_____8. Rubbing your hands together

_____9. Gasoline

_____10. Food before it is eaten

_____11. Lightening

Part 5. Transformation of Energy

Directions: Use the following forms of energy to fill in the table below: **mechanical, electrical, heat, radiant, chemical, nuclear, and sound**. The first one has been done for you.

	ORIGINAL ENERGY FORM	**FINAL ENERGY FORM**
1. Electric motor	electrical	mechanical
2. A battery that runs a moving toy		
3. A solar panel on the roof of a house		
4. A person lifting a chair		

5.	A nuclear power plant		
6.	A toaster		
7.	A church bell		
8.	Gasoline powering a car		
9.	A light bulb		
10.	Photosynthesis		